

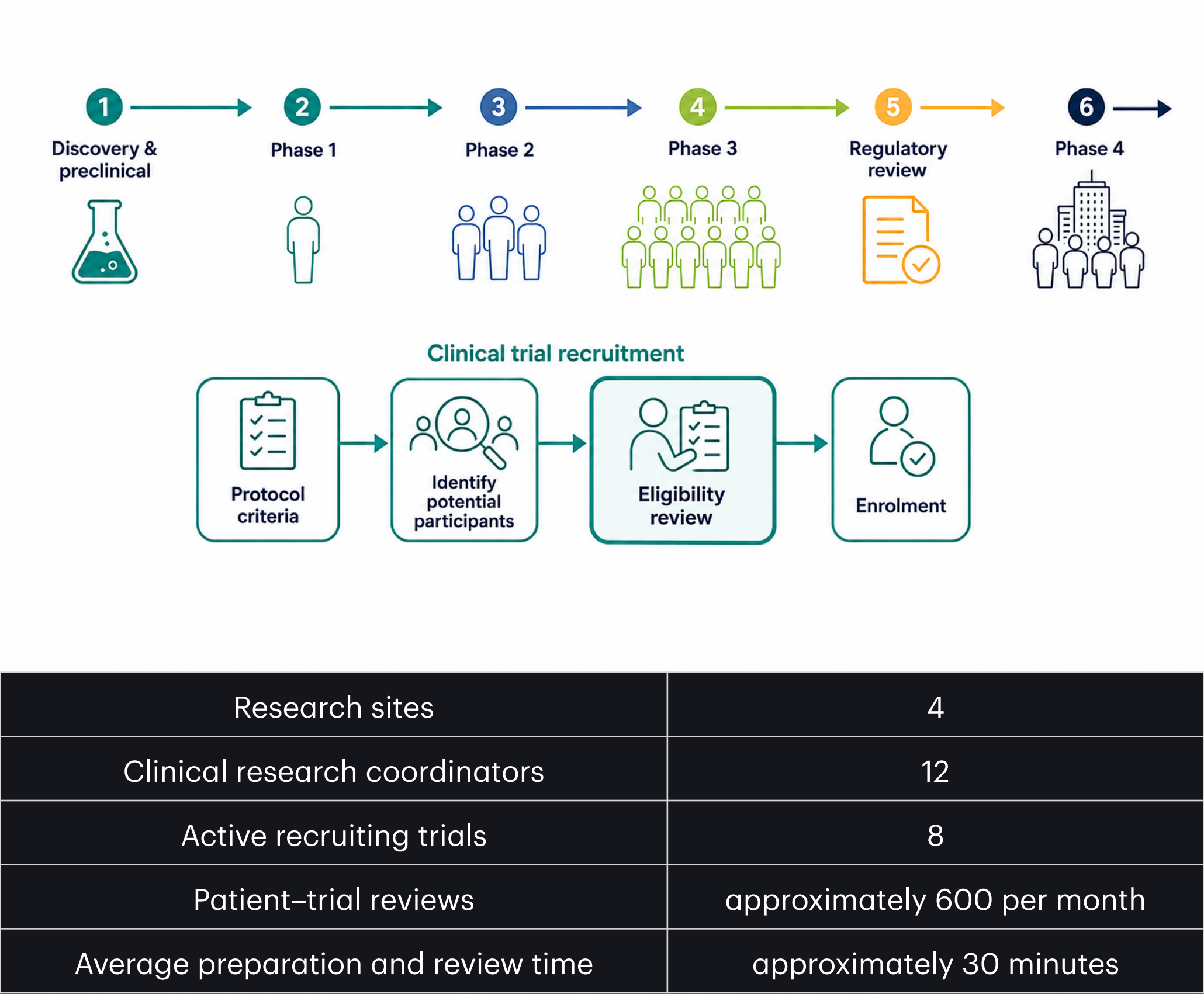
Clinical-Trial Eligibility Copilot

Human-reviewed AI support for criterion-level clinical-trial prescreening

Prepared for HelixBridge Clinical Research Network GmbH

Prescreening capacity is a bottleneck between patient identification and enrolment

- Coordinators compare narrative patient information with complex protocol criteria.
- Relevant evidence may be fragmented across EHR, CTMS and supporting documents.
- Missing information requires follow-up and professional judgement.
- Across multiple sites, manual review is difficult to scale and document consistently
- HelixBridge - **300 coordinator hours each month**



A missed candidate delays recruitment; an incorrect positive match creates operational and patient-safety risk

AI-assisted prescreening can release coordinator capacity

► when integrated into human review

Market activity	Site capacity pressure	Real implementation evidence	Nurse review time
65,004 recruiting studies	63% of surveyed sites reported hiring or retention difficulties	34% reduction in screening time and 11.1% more patients enrolled	≈12 min/match
ClinicalTrials.gov, July 2026	Global site survey, mainly US-based	12-month prospective ACTES deployment (CRC)	External 16-site AI-enabled screening example

Better coordinator decision support - not automated eligibility

AI evidence

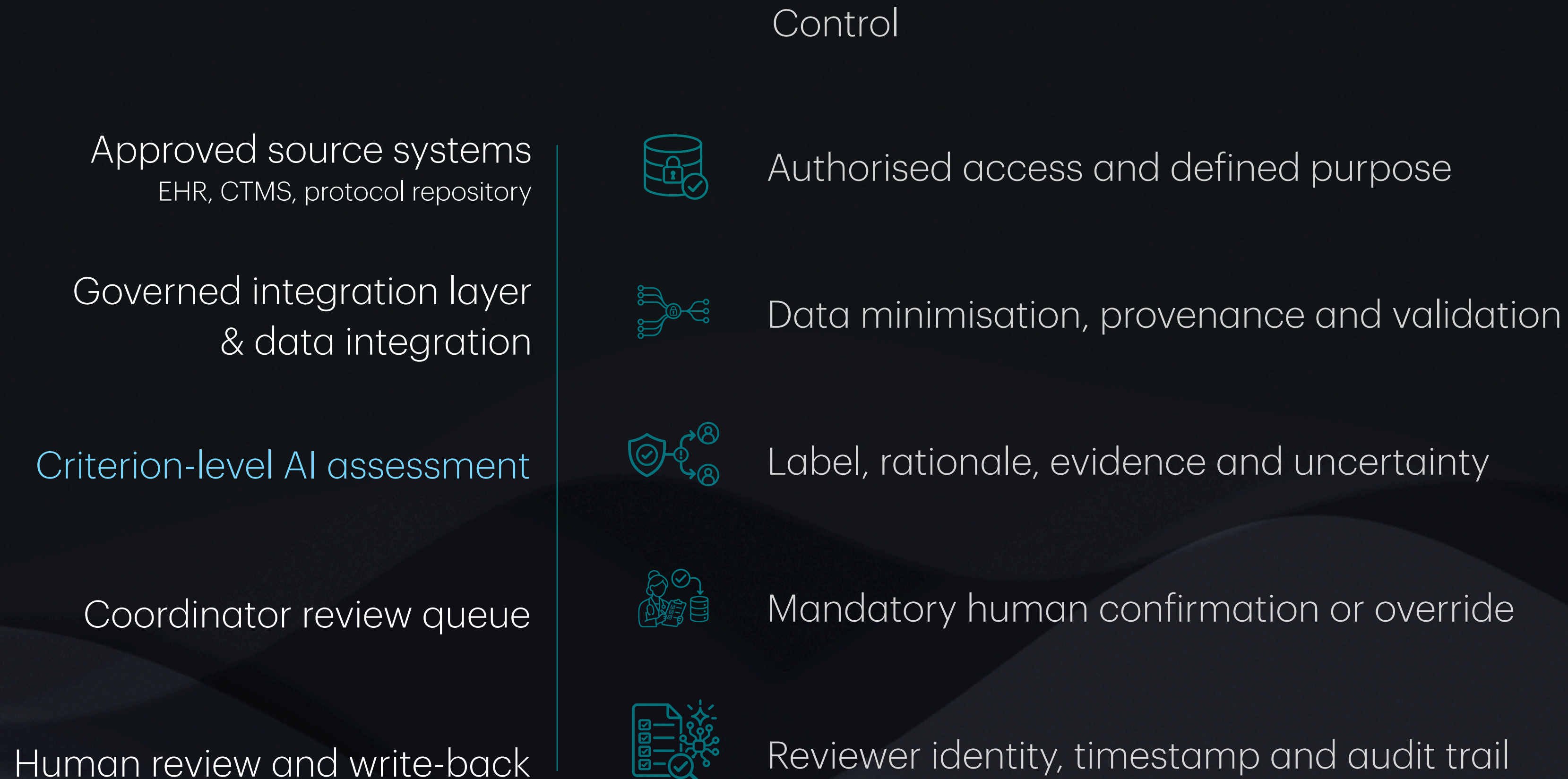
Evidence of potential	Evidence of caution
TrialGPT reported 87.3% accuracy	Evaluation used controlled research data
TrialGPT reported 42.6% shorter screening time	Results cannot be transferred directly to another workflow

The business value must be demonstrated using the network’s own workload, timing, cost and recruitment data

External evidence supports a controlled pilot - not a production-performance promise

AI supports the existing workflow - faster & safer

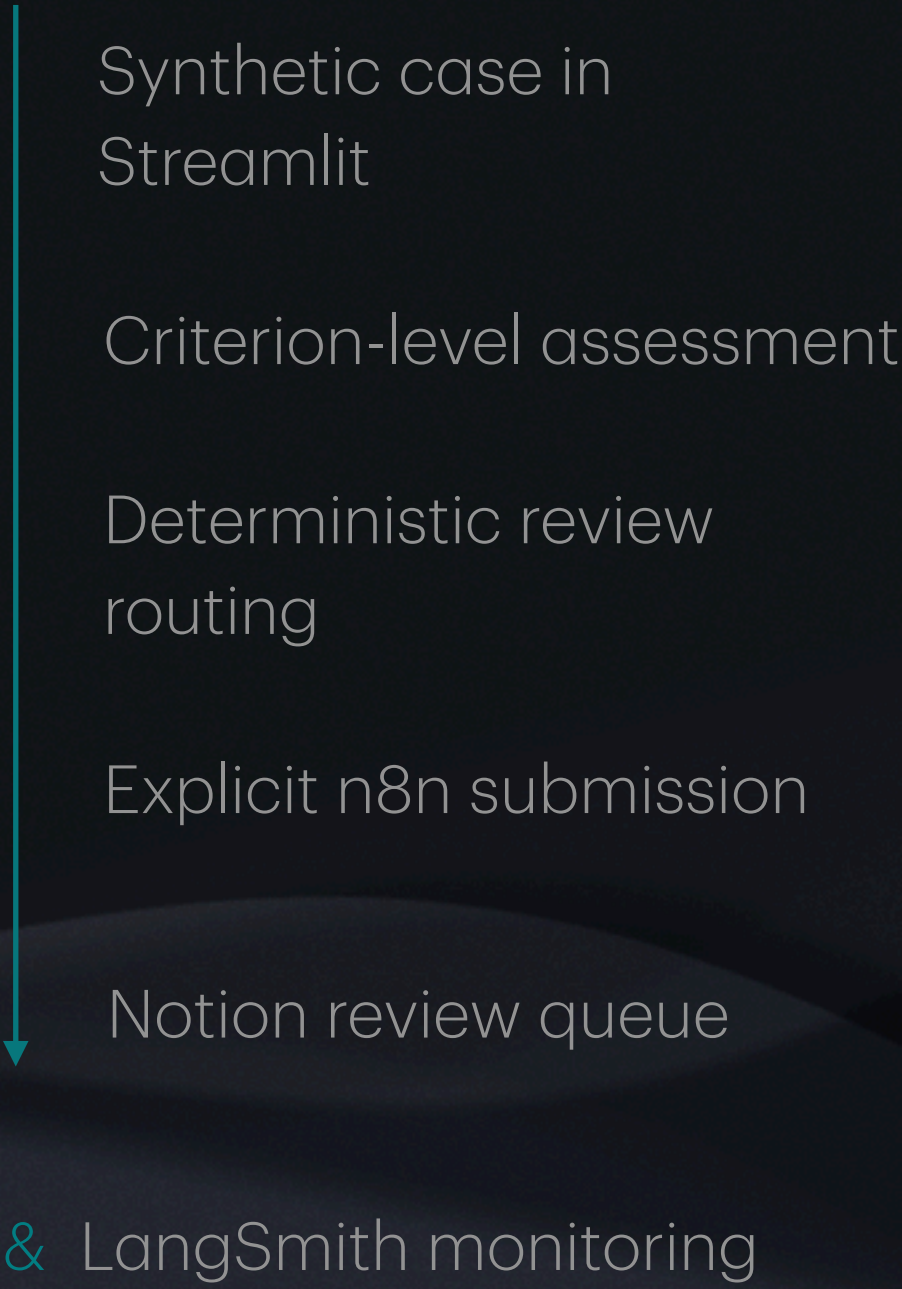
Human review by design



Authorised access · Data minimisation · Human confirmation · Audit trail · Manual fallback

MVP & demo

The complete assessment-to-review handoff



MVP placeholders	Integrated product
Synthetic Streamlit input	Governed EHR/CTMS integration
Notion review queue	Existing CTMS or validated operational work queue
Local user session	Enterprise identity and role-based access
Demonstration logging	Approved audit, monitoring and retention controls

Clinical-Trial Eligibility Copilot

Demonstration with public synthetic data only. Not for clinical or enrolment decisions.

Screening mode

Dataset case

Custom synthetic case

Patient ID

sigir-201425

Trial ID

NCT00282269

Criterion ID

CRIT_042

Synthetic patient summary

0. An 8-year-old boy fell from his bike striking his left temple on the pavement.

1. There was no immediate loss of consciousness, and a brief examination at the scene noted his pupils were symmetrical, reactive to the light, and he was moving all four limbs.

2. Half an hour after the fall the child became drowsy, pale, and vomited.

3. He was transferred to the emergency department.

4. Upon arrival the heart rate was 52/min, blood pressure of 155/98.

5. The Glasgow Coma Scale (GCS) was 6/15, the pupils were asymmetrical and movement of the right upper and lower extremities was impaired.

6. The neurosurgical team advised deferring the CT scan in favor of initiating immediate treatment.

7. The patient will provide informed consent, and will comply with the trial protocol without any practical issues.

Selected criterion

Type: exclusion

are not randomized by 6 hours after injury

Run screening

Screening result

Predicted label: UNKNOWN

Rationale: The patient summary does not state whether or not the patient was randomized within 6 hours after injury. There is no explicit documentation of the time of randomization or enrolment, so this criterion cannot be assessed.

Evidence sentence IDs:

Review route: HUMAN_REVIEW

Queue required: True

Human confirmation remains required before any clinical action.

Model: gpt-4.1 | Prompt: v2_abstention_rules | Latency: 1.06 seconds | Input tokens: 770 | Output tokens: 74

Screening result sent to the human-review queue.

Queue route: HUMAN_REVIEW

Queue status: OPEN

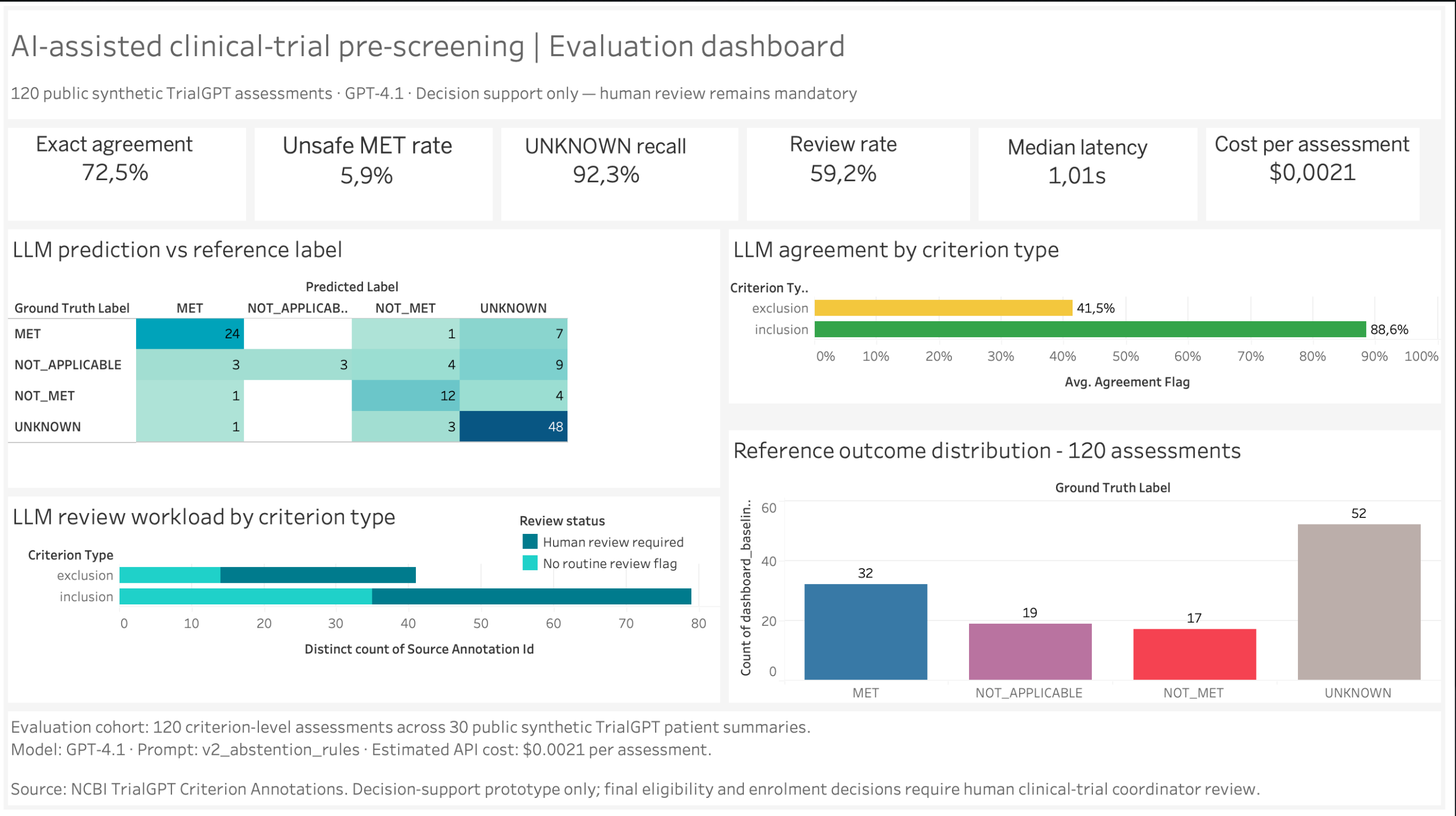
Queue ID: resp_0a817842cd82262e016a97088bc545487d2b3d84b7cf471f85c

Queue message: Synthetic assessment routed to the human-review queue.

> Evaluation-only comparison

Optimise pre-screening performance

Quality, review workload and cost



Investment should remain staged

Safety, consistency and auditability strengthen the business case

Annual workload and value assumptions

600 reviews/month × 30 minutes = 3,600 hours/year

25% net time reduction = 900 hours released/year

Quantified annual value	Amount
Coordinator capacity	40,500 €
Attributable recruitment contribution	80,000 €
Reduced rework and overflow	15,000 €
Total annual value	135,500 €

Base-case investment result

Measure	Result
Upfront all-in programme cost	160,000 €
Annual ongoing cost	60,000 €
12-month ROI	-75.0%
36-month ROI	2.7%
Indicative break-even	approximately month 35

High scenario reaches 89.3% ROI and breaks even around month 19.

59% of the projected annual value depends on attributable recruitment contribution.

The pilot must verify this assumption before full deployment.

API usage is inexpensive; clinical validation, governance, integration and human-review effort drive the real cost

Three connected risks determine whether the pilot creates value

Priority risk	Likelihood × impact	Mitigation
Incomplete source data produces misleading assessments	4 × 4 = 16 Critical	Validate required fields and provenance; flag missing evidence; route insufficient information as UNKNOWN
Reviewers over-trust AI outputs	4 × 4 = 16 Critical	Require source-first review, explicit confirmation or override, user training and sampled quality audits
Review workload offsets the expected time saving	4 × 4 = 16 Critical	Measure total preparation, correction and escalation time; monitor queue ageing; narrow or stop the pilot if net workload rises

These risks are manageable only through human confirmation, measured pilot evidence and formal stop gates.

Real-patient use requires governance before integration

Preliminary non-high-risk classification

- The system prepares criterion-level information for qualified staff.
- It does not autonomously determine eligibility, exclusion, prioritisation or enrolment.
- A qualified reviewer confirms or overrides every result.
- It is not currently intended for diagnosis, treatment or use as a medical-device safety component.

Before real-patient use

- HelixBridge Legal and Quality confirm the final EU AI Act classification.
- Medical-device qualification under the MDR is documented.
- The DPO approves the legal basis, Article 9 condition and DPIA.
- Processors, subprocessors, security controls and international transfers are approved.
- Local clinical validation, monitoring, incident response and manual fallback are operational.

Conditional approval for a readiness assessment only · not yet approved for assisted real-patient use

Any expansion in autonomy, clinical influence or intended purpose triggers a new regulatory assessment.

4-week readiness - staged consulting engagement

Engagement pathway

Readiness assessment

4 weeks · €10,000–€15,000

→

Controlled pilot

18 weeks · 1 trial · 2 sites · 4 coordinators

€35,000–€55,000

→

Conditional deployment

3–5 months · staged rollout across 4 sites

€50,000–€90,000

→

Optional operational support

Monitoring, maintenance and controlled updates

€18,000–€36,000 per year

Every phase ends with a formal STOP, PIVOT or CONTINUE decision.

Failure of a mandatory safety, privacy or human-control gate cannot be offset by financial benefit

Commercial and decision model

Primary buyer	COO or Head of Clinical Operations
Delivery model	Consulting-led implementation
Contract model	Separate fixed-price approval for each phase
Production ownership	HelixBridge controls patient data, accounts, systems and clinical decisions
Ongoing relationship	Optional monitoring, maintenance and controlled-change support

Proposed deployment gates

- Zero actions without documented human confirmation
- Zero unresolved critical safety events
- At least 25% net preparation-time reduction
- At least 80% active-user adoption
- Clinical, technical, privacy and security thresholds met

Approve the readiness assessment - not a production rollout

Readiness deliverables

- Verified workload and timing baseline
- Production data-flow and integration design
- Pilot scope, evaluation plan and acceptance thresholds
- GDPR, EU AI Act, MDR and security position
- Updated cost, risk and ROI assessment
- Final pilot recommendation and fixed scope

HelixBridge commitments

- Nominate an executive sponsor and Head of Clinical Operations.
- Nominate clinical, IT, privacy, security and quality owners.
- Select one moderate-complexity recruiting trial.
- Identify two potential pilot sites and four coordinators.
- Authorise access to aggregate workflow data and architecture information.
- Agree that readiness ends with a formal STOP, PIVOT or CONTINUE decision.

A four-week, fixed-fee readiness assessment of €10,000–€15,000

The MVP proves technical feasibility.

Readiness will determine whether a safe, useful and economically justified pilot is possible at HelixBridge

The goal is not to replace clinical judgement, but to
help experts reach safer, faster and more
transparent decisions

4-week readiness - 18-week pilot

converts assumptions into a go, pivot or stop decision

Proposed scope

1 trial · 2 sites · 4 coordinators · approximately 150 reviews per month

Stage	Duration	Evidence produced
Readiness assessment	4 weeks	Verified baseline, data assessment, compliance position and pilot charter
Build and offline validation	Pilot weeks 1–6	Integration, controls, local test set and safety evaluation
Shadow mode	Pilot weeks 7–10	Prospective evidence without influencing decisions
Assisted mode	Pilot weeks 11–16	Time, workload, adoption, corrections and escalation data
Evaluation and decision	Pilot weeks 17–18	Updated ROI, risk assessment and deployment recommendation

Failure of a mandatory safety, privacy or human-control gate cannot be offset by financial benefit

Owners and formal gates

Business value before scale

- Management cadence
- Weekly delivery meeting
 - Fortnightly clinical and risk review
 - Monthly steering committee
 - Formal STOP · PIVOT · CONTINUE gates

Decision area	Accountable client owner	Provider contribution
Funding and phase progression	Executive sponsor	Evidence, status and recommendation
Workflow and benefit realisation	Head of Clinical Operations	Process design and measurement
Clinical thresholds and residual risk	Principal investigator or medical lead	Evaluation and safety analysis
GDPR and regulatory position	DPO, legal and quality leads	Documentation and control design
Architecture and data access	Client IT and security	Integration implementation
Model, prompt and routing changes	Product owner with clinical and quality approval	Development, regression testing and release evidence

Commercial adoption signals

organisations purchase solutions in this area

- WCG reported that Deep 6 AI connected more than 28 million patients across more than 2,000 healthcare facilities in 2022.
- Tempus currently reports approximately 100 participating sites, more than 1,000 clinical locations and over 1,500 active historical trials. These are company-reported figures.
- TrialX markets recruitment, referral, trial-finder and analytics functionality to sites and sponsors

Published evidence supports controlled adoption, not guaranteed results

Evidence	Setting	Reported result	Limitation
ACTES	Prospective coordinator workflow, 12 months	34% less screening time; 11.1% more enrolled	Single institution and specific setting
Human + AI oncology study	Randomised retrospective evaluation, 355 patients	Accuracy improved; time unchanged	Retrospective records
TrialGPT	Synthetic and controlled evaluation	87.3% accuracy; 42.6% less screening time	Not a production workflow
Exigent/Tempus	16-site operational programme	Network-scale matching and human review	Vendor partnership; no controlled comparator

Sources:
ClinicalTrials.gov, Trends, Charts, and Maps
SCRS/Greenphire, Opportunities for Increased Research Efficiency, 2022
Jin et al., Nature Communications, 2024
Efiom-Ekaha et al., Journal of Clinical Oncology, 2024
ACTES prospective implementation study

HelixBridge’s own baseline and controlled pilot remain authoritative for the investment decision.

Recruitment attribution is the most important financial uncertainty

Assumption	Low	Base	High
Net time reduction	15 %	25 %	35 %
Annual capacity value	24,300 €	40,500 €	56,700 €
Additional or recovered enrolments	6	16	24
Contribution per enrolment	3,000 €	5,000 €	7,000 €
Recruitment contribution	18,000 €	80,000 €	168,000 €
Rework and overflow value	5,000 €	15,000 €	25,000 €
Total annual value	47,300 €	135,500 €	249,700 €

Scenario	12-month ROI	36-month ROI	Break-even
Low	-91.3%	-64.1%	No break-even at current operating cost
Base	-75.0%	2.7%	Approximately month 35
High	-54.0%	89.3%	Approximately month 19

Full deployment should proceed only when the business case is recalculated with measured workflow and trial-finance data.